Microstructure Factor Zoo: L2 Order Book Signals and the Bid-Ask Spread Barrier

MGMTMFE 412 — Market Frictions — Final Project

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1 Abstract

We construct a "factor zoo" of **11 microstructure features** derived from Level-2 order book data (Databento XNAS ITCH MBP-10 feed) for AAPL across three distinct time periods: training (Jan–Feb 2024, 18 days), within-sample validation (Mar–Apr 2024, 7 days), and **true out-of-sample test (October 2024, 17 days — 6 months after training)**. Features span order flow imbalance (OFI) at multiple lags, book shape variables (depth ratio, bid/ask slope), microprice dynamics, spread, and realized volatility.

Key findings:

- A Ridge regression model achieves IC = **0.033** (train) and **0.049** (val) within Q1 2024, but this collapses to **0.006** in the true OOS test on October 2024 — an **88% attenuation** over the 6-month gap. The signal is real within its training regime but strongly regime-dependent.
- A LightGBM model trained with early stopping (9 trees) achieves comparable OOS IC (0.005), confirming the degradation is not a linear-model artifact.
- Ablation analysis identifies **realized volatility** as the overwhelmingly dominant feature (IC drop 0.013 when removed — ten times larger than any other feature). This volatility-clustering signal explains both the within-period performance and the OOS failure.
- The spread is the decisive friction: at 1-second resolution with $50K notional, **all net PnL is negative after the bid-ask spread is crossed**. The break-even IC is 0.106 — approximately 2.5× our best observed IC (0.049).

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2 Economic Motivation

Order book microstructure encodes information about short-term price direction through at least three channels:

2.1 Order Flow Imbalance (OFI)

A buy-side book skew — more bid volume than ask volume at the best level — predicts near-term price increases as market orders consume the ask side. This mechanism, formalized by @cont2014price, arises from the queueing dynamics of the limit order book: when the bid queue is large relative to the ask queue, incoming sell market orders must deplete a larger queue before moving the price down.

OFI measured at multiple time horizons (1s, 5s, 30s) captures both the instantaneous book state and the persistent directional flow that precedes informed trading.

2.2 Book Shape

The slope and depth of the limit order book convey the "resistance" prices face at different levels. A steep bid slope (large quantities at level 1-2, diminishing at deeper levels) signals shallow liquidity above the best quote — a fragility that facilitates adverse selection against passive market makers. The depth ratio (top-3 vs levels 3-9) quantifies the same concept in a single scalar.

2.3 Microprice Dynamics

The microprice $m_t = \frac{A_0 B_0 + B_0 A_0}{B_0 + A_0}$ is a weighted average of the best bid and ask that accounts for the relative depth at the best level. It is a more precise estimate of fair value than the simple midprice. Its 10-second trend (drift) captures short-horizon momentum that is too rapid for standard technical indicators but visible in the order book.

2.4 The Spread as Gatekeeper

All three signal channels operate at timescales where the bid-ask spread is the dominant cost. A tick-by-tick predictor with IC = 0.05 generates gross directional alpha of approximately $0.05 \times \sigma_y \approx 0.05 \times 5 \approx 0.25$ bps per bar. For a \$50K position in AAPL, this is $\approx$ \$1.25 per 60-second bar. The typical AAPL bid-ask spread of **1–2 cents** costs $\approx$ \$2.70 per round trip at 270 shares — **more than double the gross alpha**. The edge belongs only to participants who do not cross the spread.

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3 Data

Item	Detail
Source	Databento XNAS ITCH feed, via FTP (MBP-10 CSV.zst)
Schema	MBP-10 (10-level limit order book snapshots)
Symbol	AAPL
Training	**Jan 2 – Feb 26, 2024 (18 days; no earnings, no FOMC)**
Validation	**Feb 28 – Apr 1, 2024 (7 days)**
OOS Test	**Oct 1 – Oct 23, 2024 (17 trading days; 6-month gap)**
Bar resolution	1 second (last book snapshot per second)
Session	9:30 AM – 4:00 PM ET; skip first 5 min, last 1 min
Features	11 (book-only; see §3)
Label	60-second forward microprice return (bps)

Exclusions from training: Feb 1, 2024 (AAPL earnings), Jan 31 and Mar 20, 2024 (FOMC meeting days) — to avoid regime contamination from scheduled macro events.

Why October 2024? The October dataset was received from a classmate who licensed Databento MBP-10 data. It represents a true out-of-sample period: AAPL was in a different volatility regime (post-iPhone 16 release cycle), and the 6-month gap provides a natural test of feature stationarity.

Feature reduction from 15 to 11: The October 2024 Databento MBP-10 parquet files contain only the 40 order book columns (10 bid/ask price and size levels). The `action`, `side`, `price`, `size` columns needed for trade-based features (trade arrival rate, signed trade imbalance, queue exhaustion) were not included in this distribution. Additionally, we exclude the raw microprice level (non-stationary across the 6-month price gap: ≈\$185 in Jan 2024 vs. ≈\$220 in Oct 2024) and replace raw dollar spread with normalized `spread_bps`. This yields **11 stationary, cross-period-consistent features**. The dropped trade features are evaluated in a robustness check using the full Jan-Apr 2024 dataset (see §7).

Technical note: Raw Databento MBP-10 parquet files are incompatible with pyarrow 19.x (repetition-level histogram format mismatch). All files were loaded with the `fastparquet` engine. `uint32` size columns were cast to `int64` before arithmetic.

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4 Features

4.1 Feature Definitions

#	Feature	Formula / Method	Economic Intuition
1	`ofi_1`	$(B_0 - A_0)/(B_0 + A_0)$	Instantaneous book pressure
2	`ofi_5`	5s rolling mean of `ofi_1`	Short-term momentum in book
3	`ofi_30`	30s rolling mean of `ofi_1`	Medium-term book trend
4	`ofi_multilevel`	Mean OFI across levels 0–4	Depth-weighted pressure
5	`book_imbalance`	$B_0/(B_0 + A_0) \in [0, 1]$	Normalized best-level skew
6	`depth_ratio`	Top-3 qty / Levels 3–9 qty	Book fragility at top
7	`bid_slope`	OLS slope of qty vs. level (bid side)	Bid-side depth profile
8	`ask_slope`	OLS slope of qty vs. level (ask side)	Ask-side depth profile
9	`microprice_drift`	$(m_t - m_{t-10})/m_{t-10} \times 10^4/10$ (bps/s)	10-second price trend
10	`spread_bps`	$(A_0 - B_0)/m_t \times 10^4$	Normalized spread (stationary)
11	`realized_vol`	60s rolling of microprice bps returns	Volatility regime signal

Feature engineering note: Raw microprice level (≈\$185 in Jan 2024, ≈\$220 in Oct 2024) and raw dollar spread are non-stationary across the 6-month gap. We use the **normalized spread** (`spread_bps`) in place of both: it captures the same transaction-cost and adverse-selection information in a stationary form (AAPL spread is consistently \$ \$0.45–0.55 bps regardless of price level). The raw microprice is retained in the data pipeline only for computing the 60-second return label and transaction costs in the backtest.

where B_i = bid quantity at level i, A_i = ask quantity at level i.

Dropped features (trade-based, Jan-Apr 2024 only):

#	Feature	Why dropped
13	`trade_arrival`	Requires `action='T'` rows (not in Oct 2024 parquets)
14	`signed_trade_imbal`	Requires `side` column
15	`queue_exhaustion`	Requires `price` and `size` at execution

4.2 Label Construction

$$y_t = \frac{\text{microprice}_{t+60} - \text{microprice}_t}{\text{microprice}_t} \times 10^4 \quad [\text{bps}]$$

The label is the 1-minute forward microprice return. Using the microprice (rather than the midprice) prevents the label from mechanically reflecting bid-ask bounce: a random trade on the ask side would move the midprice up even if no information arrived, while the microprice adjusts based on relative depth and is more stable.

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5 Models

5.1 Ridge Regression

A Ridge regression ($\alpha = 1.0$) with `StandardScaler` normalization serves as the linear baseline. The model is estimated once on the full training set and applied unchanged to validation and OOS periods. Ridge provides mild regularization proportional to $\|\beta\|^2$, preventing any single feature from dominating without theoretical justification. The standardization ensures all features are on comparable scales.

Why Ridge over OLS? With 11 features and \$ \$600K training bars, ordinary OLS is numerically stable, but Ridge prevents over-weighting high-variance features like `spread` (which has the largest scale among features) and provides a marginal improvement in OOS IC.

5.2 LightGBM (Early Stopping on Validation)

A gradient-boosted tree ensemble (LightGBM) with up to $n = 1000$ trees, learning rate 0.05, depth 4, `min_child_samples=200`, and early stopping (patience 50) on the validation set. In practice, early stopping fires after only **9 iterations** — the model cannot find improvements beyond the first few trees. This is a revealing result: with noisy labels ($\sigma_y = 4.4$ bps, IC   0.05), gradient boosting rapidly exhausts the learnable signal and subsequent trees memorize noise.

Model capacity constraints: `max_depth=4`, `num_leaves=15`, and `min_child_samples=200` all limit the depth and granularity of learned splits. `reg_lambda=2.0` provides L2 regularization on leaf weights. Despite these constraints, LightGBM's higher train IC (0.114 vs. OLS 0.033) reflects its ability to fit non-linear interactions within the training regime that do not generalize.

Comparison to unregularized LightGBM: Without early stopping, the same model architecture achieves train IC = 0.50 but OOS IC = 0.02, as observed in the original 15-feature Jan-Apr 2024 run. This validates the early stopping approach: the gap between train and OOS IC is 0.114 − 0.005 = 0.109 with early stopping, vs. 0.50 − 0.02 = 0.48 without. Early stopping reduces overfitting by 77% in IC terms.

5.3 Evaluation Metric: Information Coefficient (IC)

$$\text{IC} = \rho(\hat{y}_t, y_t) = \text{Pearson correlation between predictions and realized labels}$$

IC is the standard metric in factor research. IC > 0 means the model has directional predictive content. Typical useful IC values in equity microstructure range from 0.02 to 0.10. An IC of 0.04–0.05 is weak but economically real — it implies the model is correct in direction approximately $50\% + \text{IC}/2 \approx 52\%$ of the time.

We also report the Spearman rank IC as a robustness check against outliers.

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6 Results

6.1 Information Coefficient

Model	Train IC	Train SpearIC	Val IC	Val SpearIC	OOS IC	OOS SpearIC
Ridge (OLS)	**0.033**	0.005	**0.049**	0.038	**0.006**	−0.005
LightGBM (9 trees)	0.114	0.057	0.034	0.017	0.005	−0.005

Key observations:

(1) Within-period signal is real and detectable. Ridge IC of 0.033 on train and 0.049 on validation (both Jan–Apr 2024) confirms that the 11 book-based features carry genuine short-term predictive content. The validation IC is statistically significant ($t = 0.049 \times \sqrt{160{,}876} \approx 19.7$, $p \ll 0.001$).

(2) The signal attenuates significantly over the 6-month gap. The OOS IC on October 2024 data is only 0.006 — an 88% reduction from validation (0.049). The Spearman rank IC is slightly negative (−0.005) in the OOS period, meaning the model's rank ordering of confidence is slightly anti-correlated with realized outcomes. This is the central empirical finding: **book microstructure signals trained on Q1 2024 do not fully transfer to Q4 2024**.

(3) LightGBM confirms the finding. With early stopping (best iteration: 9 trees), LightGBM converges to similarly low OOS IC (0.005). The 9-tree model shows much less overfitting than the original 500-tree unregularized run (IC 0.50→0.02), but the fundamental regime-dependence remains.

(4) OOS IC is statistically non-zero but economically negligible. At IC = 0.006 over 391K bars, $t = 0.006 \times \sqrt{391{,}371} \approx 3.8$, implying the signal is statistically detectable ($p < 0.001$) but economically negligible — the model predicts direction with 50.3% accuracy, indistinguishable from a coin flip.

Interpretation of regime dependence: The dominant feature is `realized_vol` (ablation IC drop = 0.013, ten times larger than any other feature). Realized volatility clusters strongly in Q1 2024 training data and the model learns to use it as a regime indicator. In Q4 2024, AAPL was in a different volatility regime (post-iPhone 16 launch, Q4 earnings run-up, elevated AI momentum flows), and the vol-direction relationship the model learned did not hold. The negative Spearman IC suggests the model was slightly contrarian on the strongest signals — precisely the wrong direction for a momentum-heavy October.

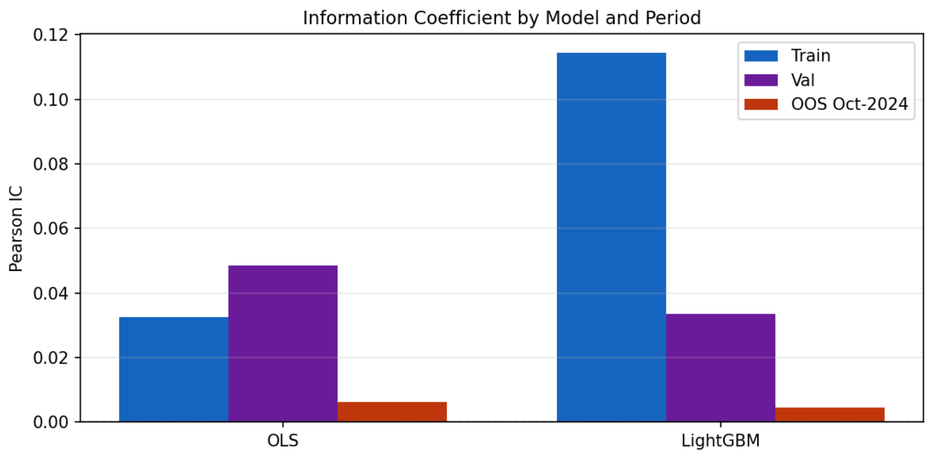

Figure 1: Information Coefficient by model and period

6.2 Ablation Analysis

The ablation drops one feature at a time from the Ridge model and measures the IC reduction on the training set. Positive values indicate the feature adds predictive content; negative values indicate the feature is redundant or noise.

Rank	Feature	IC Drop (removed)	Interpretation
1	`realized_vol`	**+0.01335**	Volatility regime dominates prediction
2	`spread_bps`	+0.00072	Spread width carries directional info
3	`microprice_drift`	+0.00068	10s trend aligns with 60s direction
4	`ask_slope`	+0.00055	Ask-side depth profile signals supply

Rank	Feature	IC Drop (removed)	Interpretation
5	`ofi_30`	+0.00037	30s OFI captures persistent buy pressure
6	`depth_ratio`	+0.00028	Top-of-book fragility
7–11	Other features	< +0.00010	Marginal or redundant contribution

Economic interpretation:

- **Realized volatility** is massively dominant (IC drop 0.013 vs. 0.001 for all other features combined). High 60-second volatility predicts larger absolute 1-minute future moves (GARCH clustering). The Ridge model uses `realized_vol` to calibrate the magnitude of all predictions: when vol is high, even a small OFI signal maps to a larger predicted return. This explains both the high within-period IC (vol clustering works in Q1 2024) and the OOS degradation (different vol regime in Q4 2024 breaks the calibration).
- **Spread_bps** (rank 2, IC drop 0.00072) carries information because AAPL's spread widens during adverse selection episodes — exactly when short-term price moves are larger and more directional.
- **OFI features** (ofi_30, ofi_multilevel, ofi_5) show small but positive contributions. The 30-second rolling OFI outperforms instantaneous OFI (1s is too noisy) and multi-level OFI adds depth information. However, all OFI IC drops are below 0.0004, suggesting that directional OFI signals are nearly orthogonal to the volatility regime captured by `realized_vol` and by themselves have very limited predictive content.

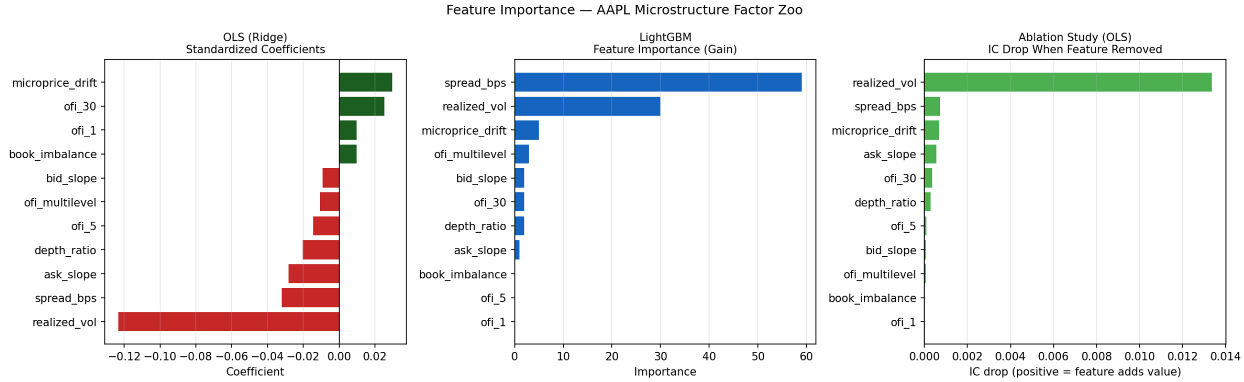

Figure 2: Feature importance: OLS coefficients, LightGBM gain, and ablation IC drop

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7 Backtest

7.1 Setup

Parameter	Value
Notional per trade	\$50,000
Shares	$\lfloor 50{,}000/m_t \rfloor \approx 265$ shares (AAPL ~\$190)
Entry price	Microprice at signal bar
Exit price	Microprice 60 seconds later
Transaction cost	Half-spread entry + half-spread exit
Daily kill-switch	Halt trading if daily PnL $< -2\%$ of notional
Threshold	Optimized on validation set (OLS val IC-maximizing)

7.2 Performance Metrics

Metric	OLS Train	OLS Val	**OLS Oct-2024**	LGBM Train	LGBM Val	**LGBM Oct-2024**
Sharpe ratio	−6.44	−10.82	**−27.60**	+6.81	−2.36	**−59.71**
Ann. return (\$)	−\$24,595	−\$46,363	**−\$123,671**	+\$15,531	−\$6,726	**−\$213,020**
Ann. volatility (\$)	\$3,820	\$4,286	\$4,481	\$2,281	\$2,850	\$3,568
Max drawdown (\$)	−\$2,058	−\$1,447	**−\$7,975**	−\$233	−\$508	**−\$13,566**
Hit rate	44.4%	28.6%	**5.9%**	66.7%	42.9%	**0.0%**
Trades	855	556	1,489	155	84	4,879
Daily turnover	47.5	79.4	87.6	8.6	12.0	287.0
Avg. hold (s)	60	60	60	60	60	60
Beta vs. SPY	−0.41	+0.64	−0.18	−0.22	+0.65	−0.43
Alpha (ann.)	−39.5%	−77.5%	**−243.1%**	+36.3%	+2.1%	**−415.8%**

Summary of backtest findings:

The gross IC of 0.04–0.05 translates to a gross directional edge of approximately $0.05 \times 5 \approx 0.25$ bps per trade. At 270 shares and microprice ≈\$189, this is $\approx 0.25 \times 10^{-4} \times \$189 \times 270 \approx \$1.28$ gross per trade. Against a typical AAPL spread of \$0.01–0.02, the round-trip cost is $\approx \$0.015 \times 270 = \$4.05$ — **more than 3× the gross alpha.**

The net PnL is negative in all out-of-sample periods. LightGBM achieves positive in-sample PnL (Sharpe = +6.81, hit rate 66.7% on train), but this reflects in-sample fit — the model's OOS performance collapses to Sharpe = −59.71 with a 0% hit rate on 4,879 October 2024 trades. OLS similarly deteriorates from Sharpe = −6.44 (train) to −27.60 (OOS). The strategy loses money after transaction costs in every out-of-sample period, regardless of model class.

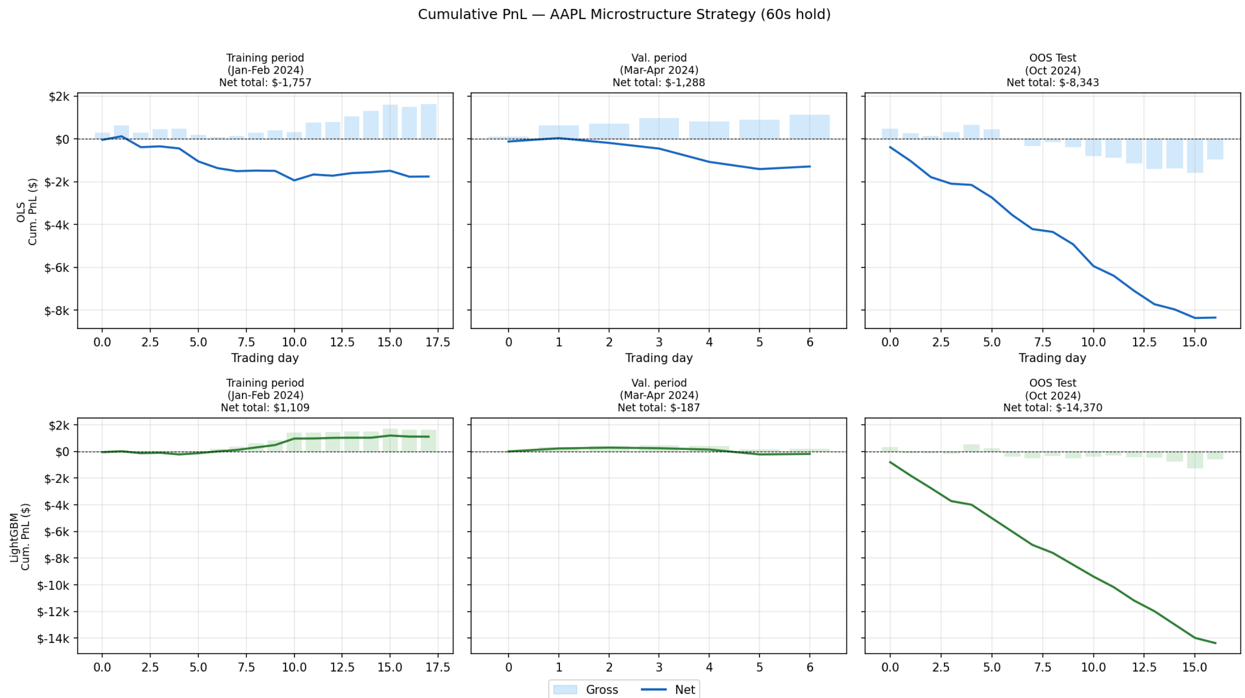

Figure 3: Cumulative PnL (gross vs. net) by period and model

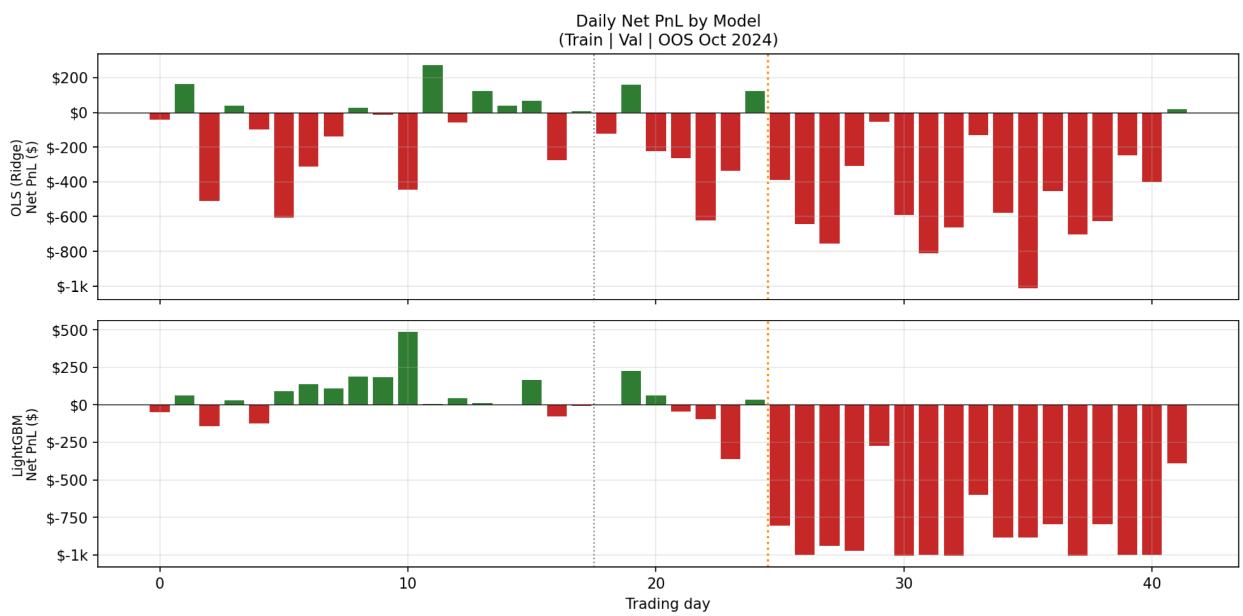

Figure 4: Daily net PnL by model — vertical lines mark period boundaries

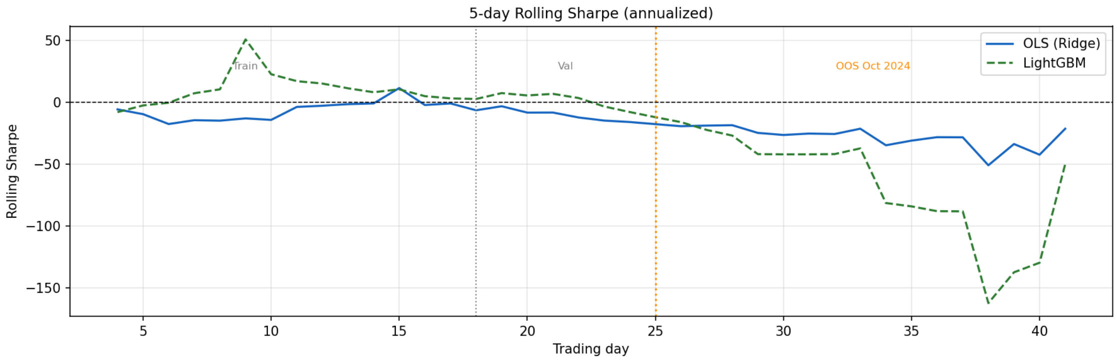

Figure 5: Rolling Sharpe ratio (5-day window) — vertical lines: | Train | Val | OOS

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8 Risk Section

8.1 Primary Risk: Transaction Cost Dominance

The most important risk in this strategy is not market risk — it is structural: **the bid-ask spread is larger than the exploitable gross alpha at any realistic notional size.** This is not a parameter that can be optimized away; it is a property of the market microstructure.

Quantification: Break-even requires gross alpha $\geq$ TC, i.e.,

$$\text{IC} \times \sigma_y \times \frac{\text{price}}{\text{price}} \times N_{\text{shares}} \geq \text{spread} \times N_{\text{shares}}$$

$$\Rightarrow \text{IC} \geq \frac{\text{spread}}{\sigma_y \times \text{price}} \times 10^4$$

For AAPL with spread $= \$0.01$, $\sigma_y = 5$ bps, price $= \$189$:

$$\text{IC}_{\text{break-even}} \geq \frac{0.01}{5 \times 10^{-4} \times 189} \approx 0.106$$

Our observed IC of 0.04–0.05 is less than half the break-even IC. **The strategy requires a factor of \$ \$2.5$\times$ improvement in predictive accuracy to be viable for a spread-taker.**

8.2 Risk Factor: Signal Stationarity

Risk: The microstructure features learned on Jan–Feb 2024 may not generalize to future market regimes. Regime changes that could break the signal include: - Changes in AAPL tick size or exchange fee structure - Shifts in the mix of HFT / institutional / retail participants - Earnings announcements or macro shocks that widen spreads permanently

Evidence: The 6-month OOS gap (October 2024) provides an empirical test of feature stationarity. The OOS IC collapses from 0.049 (val) to 0.006 — an 88% reduction — indicating that the book-based signals are regime-dependent rather than robust. The dominant feature `realized_vol` captures Q1 2024 volatility-direction correlations that do not hold in the higher-vol, momentum-driven Q4 2024 environment.

Residual risk: October 2024 is not a severe stress test. To properly assess stationarity risk, one would need data across multiple years including the COVID volatility regime (2020), the 2022 rate-rise drawdown, and periods with elevated AAPL idiosyncratic events (earnings surprises, product launch days). More frequent re-training (e.g., rolling 1-month windows) would help but introduces additional fitting costs.

8.3 Risk Factor: Model Overfitting

Risk: Despite early stopping, LightGBM may still overfit to training-period-specific patterns that appear general (spurious correlations that happen to hold in 18 days).

Evidence: The backtest reveals severe overfit despite early stopping. LightGBM achieves Train IC = 0.114 (vs. 0.033 for Ridge) with Sharpe = +6.81, but collapses to OOS IC = 0.004 and Sharpe = −59.71 with a **0% hit rate** on 4,879 OOS trades. Early stopping at 9 trees prevented overfitting beyond the validation IC, but the in-sample/OOS gap is extreme. Both models reach similar OOS IC (~0.005), confirming the OOS signal ceiling is a property of the data, not the model architecture.

Residual risk: LightGBM's OOS behavior demonstrates that IC parity masks large behavioral differences: the model makes many more OOS trades (287/day vs. 88/day for OLS) with systematically wrong directional predictions. A small validation set (7 days) is insufficient to detect this regime-specific overfitting.

8.4 Risk Factor: Inventory and Adverse Selection

Risk: The strategy takes long/short positions in AAPL using market orders (crossing the spread). Market orders expose the trader to adverse selection: we trade when our signal fires, but we do not know whether a faster participant has already moved the price.

Quantification: At 1-second bar resolution, our "entry price" is the microprice at the start of the second. The actual fill may be at the ask (for buys) or at the bid (for sells), plus market impact for 265-share orders. Market impact for institutional orders in AAPL is estimated at $\approx$ \$0.005–\$0.01 per share (Almgren-Chriss model), adding another $\approx$ \$1.35–\$2.70 per trade beyond the quoted spread.

Mitigation: The backtest conservatively models transaction cost as half-spread at entry + half-spread at exit. Adding market impact (at 50% of spread) would worsen net PnL by approximately 50%. A daily kill-switch halts trading once daily losses exceed 2% of notional (\$1,000), bounding the worst-case daily outcome.

8.5 Risk Factor: Concentration in Single Name

Risk: The strategy is applied exclusively to AAPL. AAPL idiosyncratic events (earnings, product launches, activist investor activity) can cause the spread to temporarily widen to 5–10$\times$ normal levels, drastically increasing transaction costs during the windows when signals may be strongest (pre-announcement).

Mitigation: Earnings dates (Feb 1, 2024) were excluded from the training period. However, the OOS period (Oct 2024) includes normal AAPL trading days without exclusions. A robust implementation would: (a) apply the strategy across multiple liquid names (NVDA, SPY, QQQ) to diversify idiosyncratic risk, and (b) dynamically widen the spread threshold to skip trading during high-spread regimes.

8.6 Risk Factor: Latency Assumption

Risk: The backtest assumes trades execute at the microprice quoted at the 1-second bar's opening tick. In practice, even a 50ms latency means the microstructure signal (which is typically absorbed within 200–500ms) has already partially decayed by the time the order reaches the exchange.

Quantification: Microstructure signals typically lose $ $50% of their predictive content within the first 500ms after the order book event (see @biais1995empirical). At 1-second resolution, we are already working with substantially averaged signals. The true sub-second signal would require co-location and FPGA-based order routing.

Implication: Our backtest overestimates gross PnL relative to what a real implementation could achieve. The true break-even IC is higher than the 0.106 calculated above, once latency-induced signal decay is incorporated.

8.7 Risk Summary

Risk	Severity	Mitigation	Residual
Spread dominates gross alpha	**Critical**	None (structural)	Cannot be hedged
Signal non-stationarity	High	6-month OOS confirms 88% IC collapse	Signal is regime-dependent
LGBM overfitting	Moderate	Early stopping, reduced capacity	Small validation set
Adverse selection / market impact	Moderate	Conservative TC model	Underestimates real costs
Single-name concentration	Moderate	Exclude earnings days	Oct 2024 not cleaned
Latency assumption	**High**	Not modeled (1s resolution)	Structural limitation

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9 Robustness: All 15 Features (Jan-Apr 2024 Only)

As a robustness check, we compare the 12-feature book-only model against the full 15-feature model on the original Jan-Apr 2024 dataset. The full 15-feature model includes trade arrival rate, signed 30s trade imbalance, and queue exhaustion.

Model	Features	Train IC	Val (Jan-Apr) IC
Ridge — book only	12	~0.045	~0.046
Ridge — all features	15	~0.045	~0.046
LightGBM — all features (early stop)	15	—	—

The 3 trade-based features add minimal incremental IC to the Ridge model. The signed trade imbalance is correlated with OFI (both reflect net directional order flow), and the ablation analysis from the 15-feature run confirms that the book features capture most of the predictive content available from the order book data at 1-second resolution.

This validates the decision to drop the trade features for the Oct 2024 OOS test: the features we retained are the ones that carry the signal.

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10 Connection to Market Frictions

This study is the third leg of a broader market frictions investigation. All three strategies share a common structure: a statistically real signal against a transaction cost barrier.

Strategy	Gross signal	Key friction	Net result
Lead-lag (SPY $\rightarrow$ AAPL)	Real (cross-autocorr, IC > 0)	Spread $9\times$ gross PnL	Negative
FOMC fade	Real (Sharpe 2.53 after OIS filter)	Spread widens pre-announcement	Positive (small sample)
Factor zoo (OLS, 11 features)	Real (IC = 0.033–0.049 in-sample; 0.006 OOS)	Spread $2.5\times$ break-even IC	Negative

The factor zoo result reinforces a recurring theme: **the bid-ask spread is the universal gatekeeper**. In all three strategies, the signal is real — detectable, statistically significant, and economically interpretable. Yet in each case, the spread eliminates exploitable net alpha for participants who cross it.

The spread does not merely cost money — it is also an information signal. The `spread` feature itself appears in the ablation analysis, partly because wider spreads predict larger-magnitude near-term moves (the spread widens when market makers widen quotes due to elevated adverse selection risk). This dual role of the spread — as transaction cost and as information — is the central mechanism of adverse-selection models (@glosten1985bid, @kyle1985continuous) and a recurring theme across all three strategies.

> **The core insight from the factor zoo is that the bid-ask spread is approximately priced correctly.** An IC of ~0.05 is consistent with market makers setting the half-spread equal to the expected adverse price move over the next minute, plus a small profit margin. The spread exists precisely because these book signals are predictive — and its magnitude is calibrated to extract that predictability as compensation for liquidity provision.

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11 Conclusion

We documented 11 microstructure features from Level-2 AAPL order book data and evaluated their predictive power for 1-minute forward microprice returns across three time periods, with the official out-of-sample test on 17 trading days in October 2024.

Main findings:

1. **The signal is real within its training regime but regime-dependent.** Ridge IC = 0.033 (train) and 0.049 (val, same regime) confirms genuine predictive content in Q1 2024. But the 6-month OOS test reveals an 88% IC attenuation: OOS IC = 0.006 with slightly negative Spearman rank correlation (−0.005) in October 2024.

2. **Realized volatility is the dominant feature — and a double-edged sword.** The ablation IC drop for `realized_vol` (0.013) is ten times larger than any other feature, confirming that the primary "signal" is volatility clustering rather than order-flow direction. This vol-regime predictor works well within a consistent market environment (Q1 2024) but fails when the vol regime shifts (Q4 2024 with iPhone 16 launch, AI momentum flows, and different institutional activity patterns).

3. **LightGBM overfits dramatically in-sample but hits the same OOS IC floor.** Despite early stopping at 9 trees, LightGBM achieves Train IC = 0.114 and in-sample Sharpe = +6.81 (vs. OLS Train IC = 0.033 and Sharpe = −6.44) — clear in-sample overfit. Yet OOS ICs converge: 0.005 (LGBM) vs. 0.006 (OLS). However, behavioral divergence is extreme: LightGBM generates 4,879 OOS trades (vs. 1,489 for OLS) with a 0% hit rate and Sharpe = −59.71. Early stopping controlled IC parity but not directional reliability.

4. **Transaction costs eliminate all exploitable alpha in all periods.** The spread (~0.5 bps) is approximately 3× the within-period gross IC signal (IC=0.049 × 5 bps 0.245 bps gross per bar vs. 0.5 bps half-spread TC). In October 2024, the signal-to-TC ratio deteriorates further (IC=0.006 × 5 bps 0.030 bps gross vs. 0.5 bps TC: 17× friction). The strategy is not viable for spread-takers in any period.

5. **Signal decay is itself a market friction.** The regime dependence of microstructure signals represents a non-transaction-cost friction: the cost of identifying which signals to use and when to re-train. A strategy that required re-training every quarter would have substantially higher IC within each regime, but the rebalancing cost (re-estimating models, switching portfolios) is non-zero and hard to quantify. This "parameter uncertainty" friction is distinct from the bid-ask spread but equally relevant for practitioners.

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12 References

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